



Fashion Recommenders System

TABLE OF CONTENTS

01

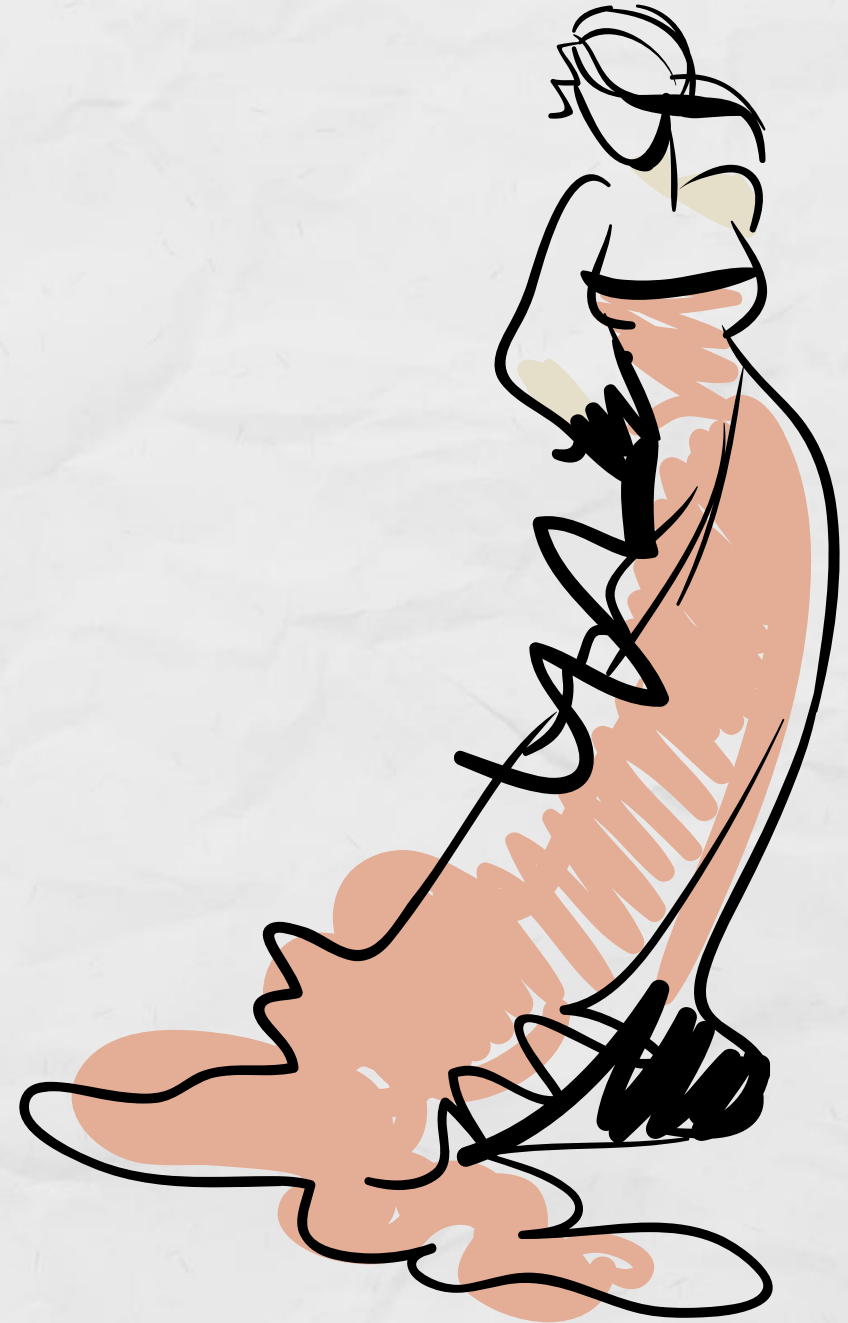
Introduction

02

Data
Preprocessing

03

Models





01 | Introduction

Abstract

- Aims to develop three advanced recommendation systems for Twenty Too[®], a company specializing in AI-as-a-service solutions for the fashion industry.
- These systems include: a frequently bought together model to suggest complementary products, a product similarity model to recommend visually and contextually similar items, and a personalization model tailored to individual user preferences based on demographics and browsing history.



Introduction

The first system, a frequently bought together model

The second system focuses on product similarity

The third system is a personalization model that tailors recommendations based on individual user demographics and browsing history.



Welcome To TwentyToo Fashion Recommendation Engine

Discover your perfect style with our curated selection of fashion-forward picks. Whether you're looking for complete looks with our 'Top Picks Together' suggestions, exploring 'Similar Styles' for trending matches, or seeking 'Personalized Picks' tailored just for you, we're here to elevate your wardrobe effortlessly. Start exploring and let your style shine!



Top Picks Together



Similar Styles



Personalized Picks

Screen Recorder is sharing your screen.

Stop sharing

[Hide](#)



02

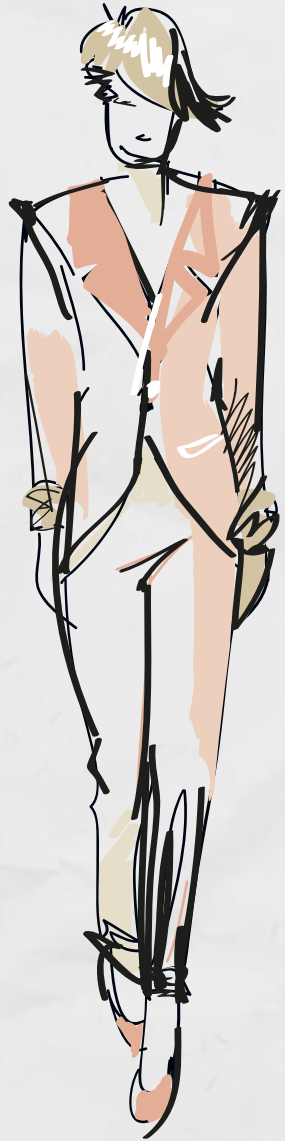
Data Preprocessing

Preprocessing 2 datasets

Data1 : users data before
1/6/2023

Data2 : users data after
1/6/2023





First Problem

Extracting Product IDs



Product_Package Before

```
'{"product_id": "ab27bf375ed51569746345ece5d0dd9e49a39768", "product_brand": "Premoda", "website": "tfk"}'
```

Product_Package After

```
ab27bf375ed51569746345ece5d0dd9e49a39768
```

. Methodology

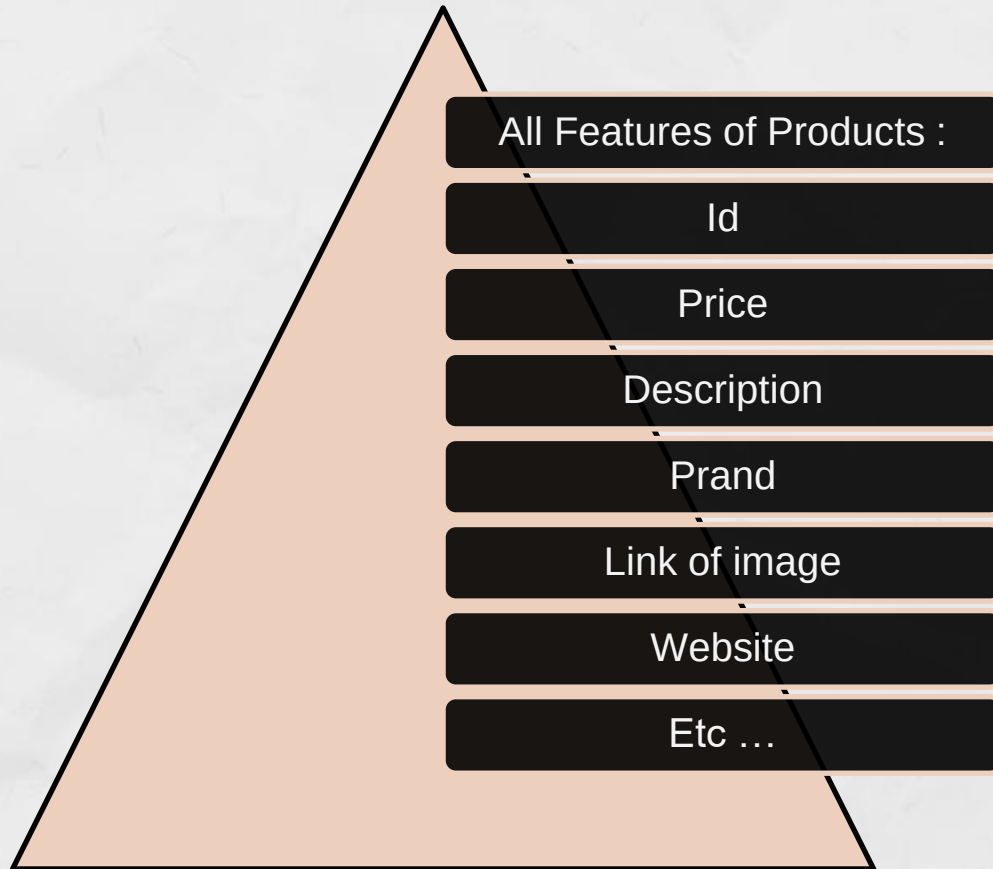
**Regular
Expressions:**

Pattern Used:
`[a-fA-F0-9]{32,}`

**Exploding
Columns**

**Data
Transformation**

Product data





Features Has multi
values



- extracts specific price information from the price column which contains lists of dictionaries in string format.
- retrieves attributes such as price value, currency, original status, valid from, and valid to dates, storing them in new columns.

```
'[{'value': 18.0, 'currency': 'USD', 'is_original': True, 'valid_from': '2023/09/13', 'valid_to': False}]'
```



Sample from Other attributes column

```
{\ '_id\': ObjectId('\663b200a012b93dcf0dee005\'), \ 'sku\': \ 'st2205178018028785\ ', \ 'Product_URL\': \ 'https://www.shein.com//GLOWMODE-FeatherFit-Retro-Sport-24-Leggings-p-11096183-ca  
t-2190.html?src_identifier=st%3D2%60sc%3Dsports%20legging%60sr%3D0%60ps%3D1&src_module=search&src_tab_page_id=page_goods_detail1672574085241&mallCode=1&scici=Search~~EditSearch~~1~~sp  
orts_20legging~~~~0\ ', \ 'All_Reviews_Rating\': \ '4.93\ ', \ 'All_Reviews_Count\': 57.0, \ 'Clothing-Color\': [\ 'Russet Orange\ '], \ 'thumbnail\': {\ 'Type\': \ 'image\ ', \ 'Name\': \ 'Russet  
Orange\ ', \ 'Value\': \ 'thumbnails/f5ba774272a55c90af9e7c388245eeca61835f2e.jpg\ '}, \ 'Images\': \ 'https://img.ltwebstatic.com/images3_pi/2023/01/11/16734228854574b7a6d68d6d5e091427a5dc  
afd2c9.jpg;https://img.ltwebstatic.com/images3_pi/2022/07/25/16587271198da8f58f0a2c820b6c0da28326048a25.jpg;https://img.ltwebstatic.com/images3_pi/2022/07/25/1658727123945915ff42bfb54  
cfbfc446c3cac4698.jpg;https://img.ltwebstatic.com/images3_pi/2022/07/25/16587271215ccb3cd3204c3edc7ec38041e90e4.jpg;https://img.ltwebstatic.com/images3_pi/2022/07/25/16587271242405  
d05f50b56b731ade77aeb05b076b.jpg;https://img.ltwebstatic.com/images3_pi/2022/07/25/1658727118c138e25c7a270770d979acd27ea1ad87.jpg\ ', \ 'CM_product_measurements\': \ 'size:XS;Length:74.8  
cm;Inseam:61.5 cm;Waist Size:58 cm;Hip Size:68.5 cm;Thigh:43.8 cm;EU:34 ; size:S;Length:76 cm;Inseam:62 cm;Waist Size:62 cm;Hip Size:72.5 cm;Thigh:46 cm;EU:36 ; size:M;Length:77.2 cm;  
Inseam:62.5 cm;Waist Size:66 cm;Hip Size:76.5 cm;Thigh:48.2 cm;EU:38 ; size:L;Length:78.4 cm;Inseam:63 cm;Waist Size:71 cm;Hip Size:81.5 cm;Thigh:51 cm;EU:40/42 ; size:XL;Length:79.6  
cm;Inseam:63.5 cm;Waist Size:76 cm;Hip Size:86.5 cm;Thigh:53.8 cm;EU:44 ; size:XXL;Length:80.8 cm;Inseam:64 cm;Waist Size:81 cm;Hip Size:91.5 cm;Thigh:56.6 cm;EU:46\ ', \ 'CM_body_measu  
rements\': \ '\ ', \ 'Availability\': True, \ 'original_thumbnail_url\': \ 'https://img.ltwebstatic.com/images3_pi/2022/07/25/1658727127bd439737fa3addb6c7f3af46064308a0.jpg\ ', \ 'Clothing-D  
epartment\': [\ 'Pants\ '], \ 'preview_image\': \ '0001d453006af5200ac06d23a1c8e30b219c4a83_0.jpg\ ', \ 'status\': [\ 'new\ '], \ 'default\': True, \ 'child_product_ids\': None, \ 'lang\': \ 'en  
\ ', \ 'Length\': \ 'Cropped\ ', \ 'Features\': \ 'Add TACTEL® Fibre\ ', \ 'Sheer\': \ 'No\ ', \ 'Clothing-Pattern\': [\ 'Plain\ '], \ 'Clothing-Detail\': [\ 'Contrast Binding\ '], \ 'Clothing-Type\':  
[\ 'Regular\ '], \ 'Clothing-Waistline\': [\ 'High Waist\ '], \ 'Clothing-Fabric-Elasticity\': [\ 'High Stretch\ '], \ 'Clothing-Material\': [\ 'Fabric\ '], \ 'Clothing-Composition\': [\ '19% Span  
dex\ '], \ 'All-Care\': \ 'Machine wash, do not dry clean\ ', \ 'Original_Description\': \ 'Color: Russet Orange; Pattern Type: Plain; Details: Contrast Binding; Type: Regular; Waist Line:  
High Waist; Length: Cropped; Features: Add TACTEL® Fibre; Fabric: High Stretch; Material: Fabric; Composition: 19% Spandex; Care Instructions: Machine wash, do not dry clean; Sheer: N  
o; \ ', \ 's_f\': [\ 'All_Reviews_Rating\ ', \ 'All_Reviews_Count\ ', \ 'Clothing-Color\ ', \ 'Clothing-Department\ ', \ 'All-Price\ ', \ 'All-Brand\ ', \ 'Clothing-Size\ ', \ 'Clothing-Pattern\ ', \ 'C  
lothing-Detail\ ', \ 'Clothing-Type\ ', \ 'Clothing-Waistline\ ', \ 'Clothing-Fabric-Elasticity\ ', \ 'Clothing-Material\ ', \ 'Clothing-Composition\ ', \ 'All-Care\ '], \ 'is_available\': True,  
\ 'product_description\': \ 'a pants with color: russet orange _ department: pants _ detail: contrast binding _ material: fabric _ pattern: plain _ type: regular _ waistline: high waist  
_ length: cropped _ sheer: no _ target_audience: women _ title: glowmode featherfit™ retro sport 24" leggings\ '}<
```

Other attributes

id	URL	Rating	Count
color	image	product measurements	original description .

Pattern : extract every feature in other attributes to individual columns `'([^'])'?: ?('^[^']*|'[^,}]*')+)`

ID Before

```
objectId( '663b200a012b93dcf0dedffe' )
```

ID After

```
663b200a012b93dcf0dedffe
```

ID Feature

- Extracts the hexadecimal part of the object ID string from the ID column. This is useful for standardizing the ID format.
- `object_id_string.split("(")[1].split(" ")[0].strip("'")`





Filtering Data



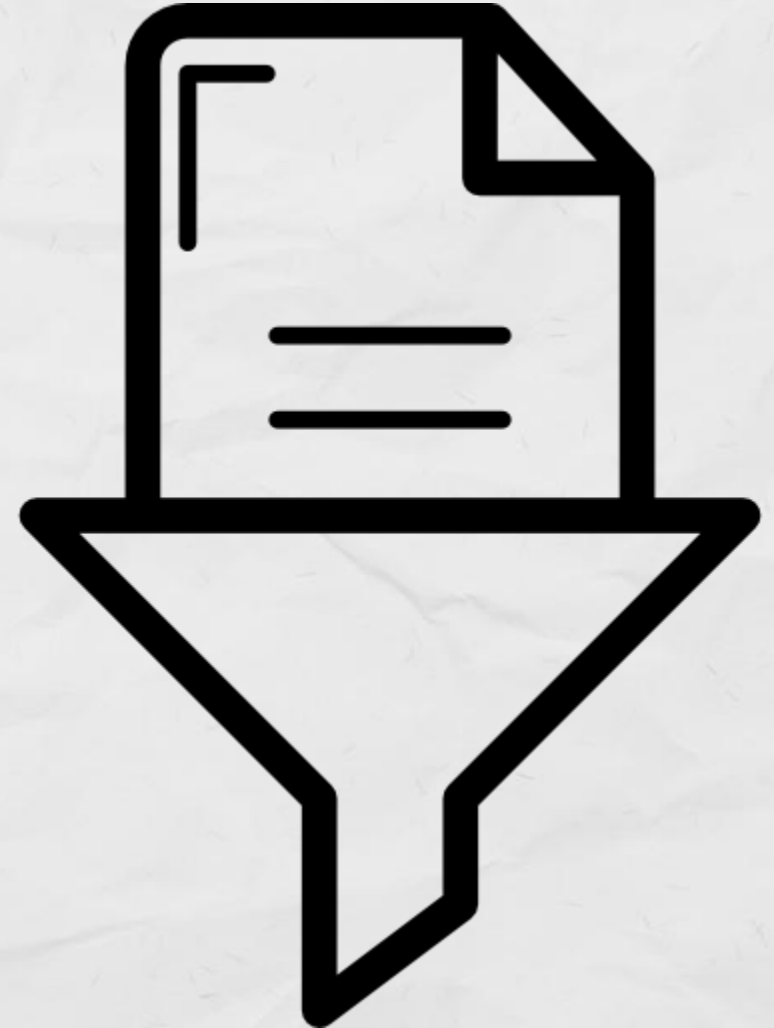
Events of Interest

- Select events that indicate significant user interactions with products

Button Names of Interest

- Include interactions with specific buttons to capture user intent and preferences

- Merge all processed into a single DataFrame.





Extract Age from 'Answer' Column



Extract Age from 'Answer' Column

Objective: Derive the age of users from their responses in the 'Answer' column.

- **Method:**

- Use regular expressions to parse birthdates from the 'Answer' column:

Pattern: '\b\d{2}-\d{2}-\d{4}\b'

- Calculate age based on the extracted birthdate.





Add Age and Age Group



Add Age and Age Group

Add Age and Age Group

- Categorize users into age groups based on their extracted ages.

Fill Missing Ages

- . Group 'User_ID' and apply forward and backward filling.
- . Use linear regression to predict missing ages based on other features



Filter Data & Apply Interaction Mapping





Handle Missing Values in 'City' and 'Region'



Handle Missing Values in 'City' and 'Region'

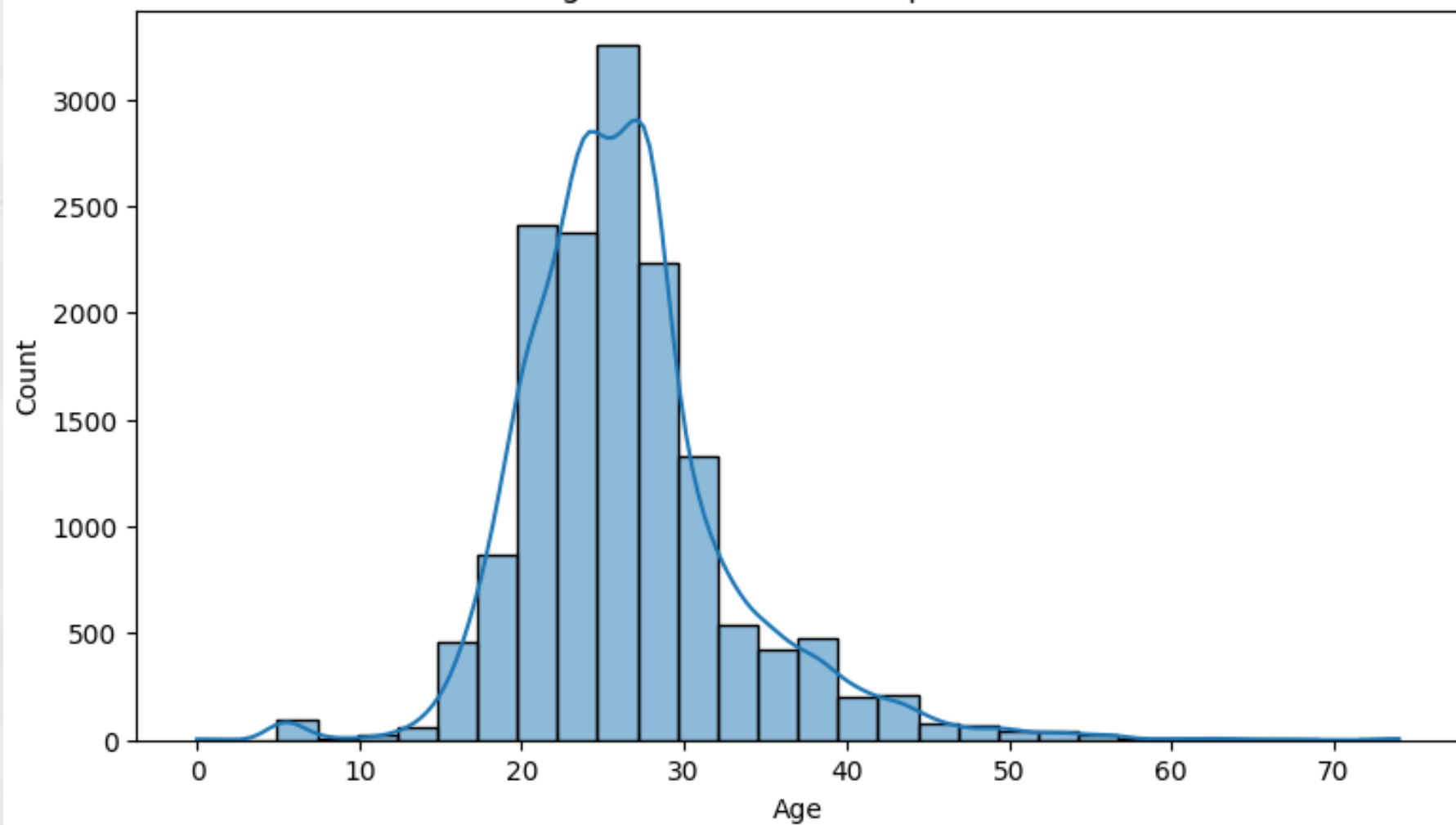
Fill missing 'region' values with corresponding 'city' values and vice versa.

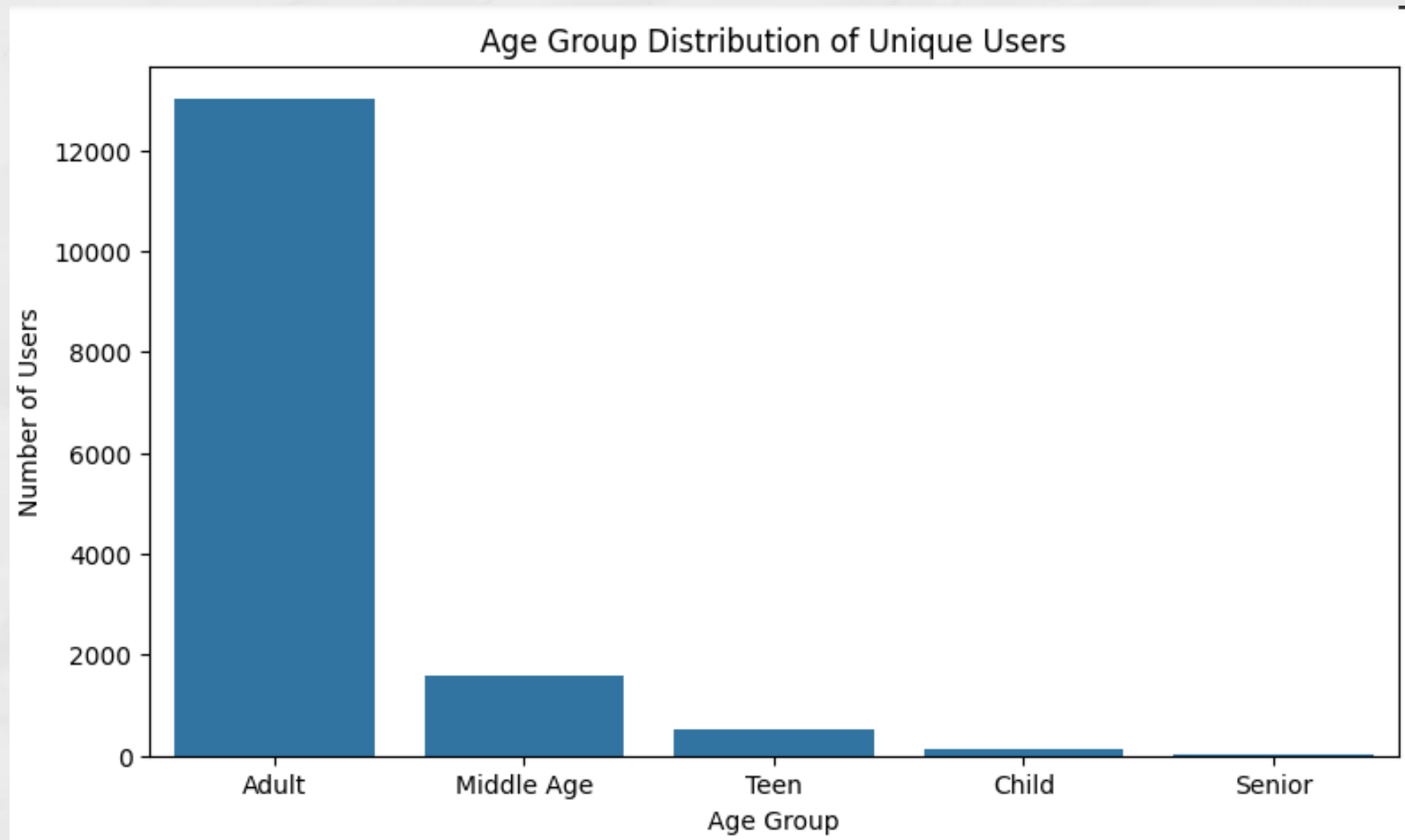
Apply forward and backward filling within user groups.

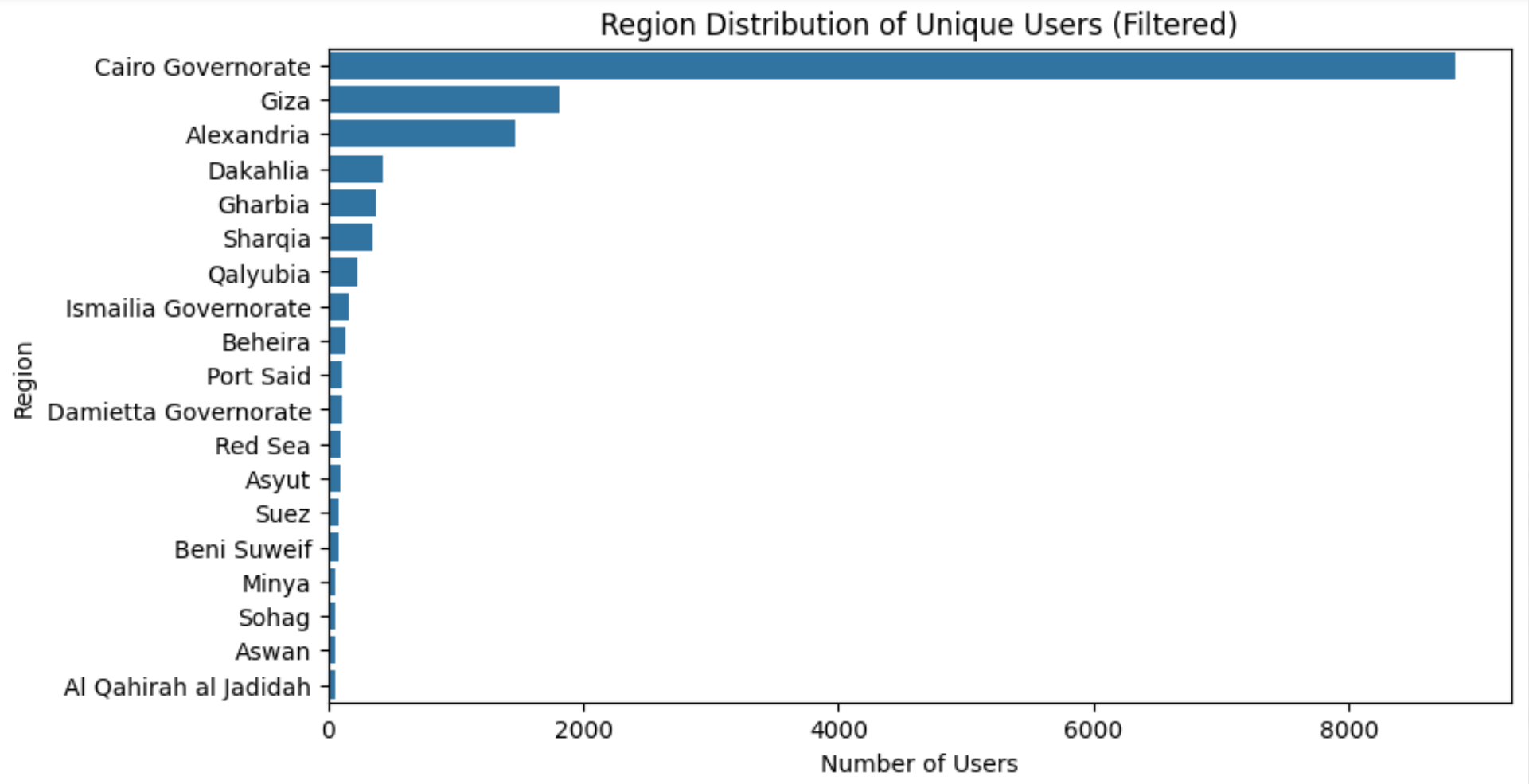
Exploratory Data Analysis



Age Distribution of Unique Users

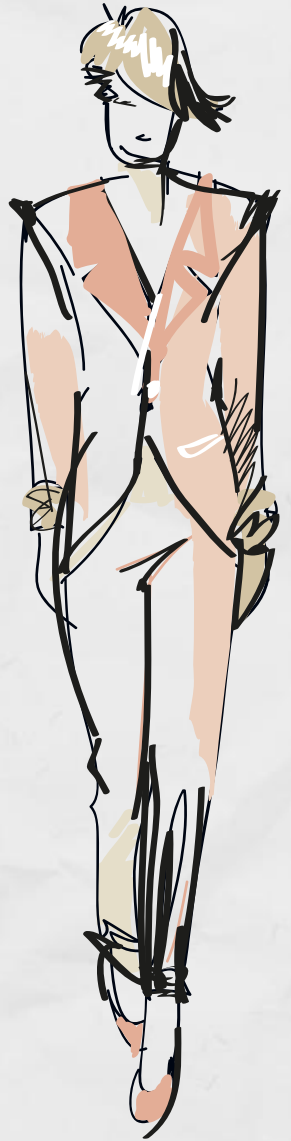








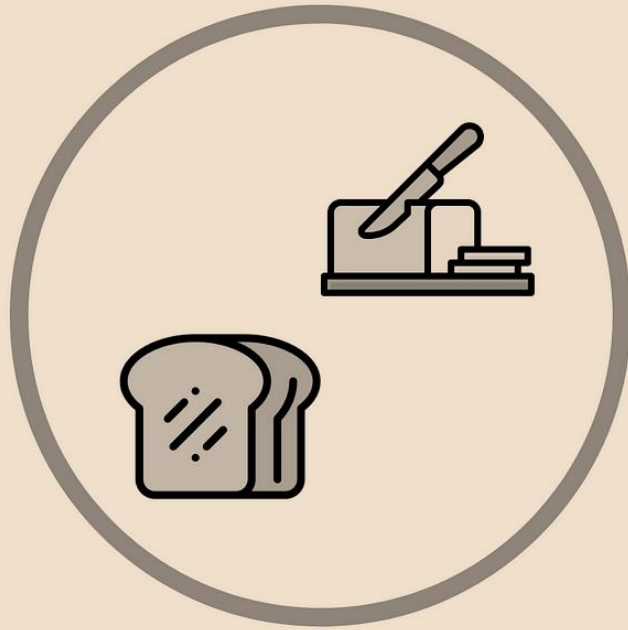
03 | Models



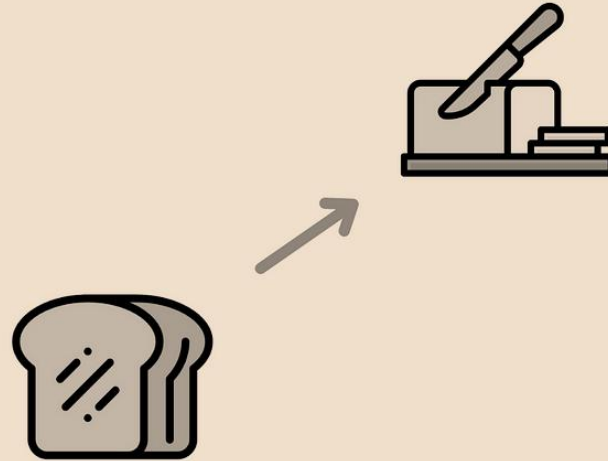
Frequently Bought Together



Association Rule Mining



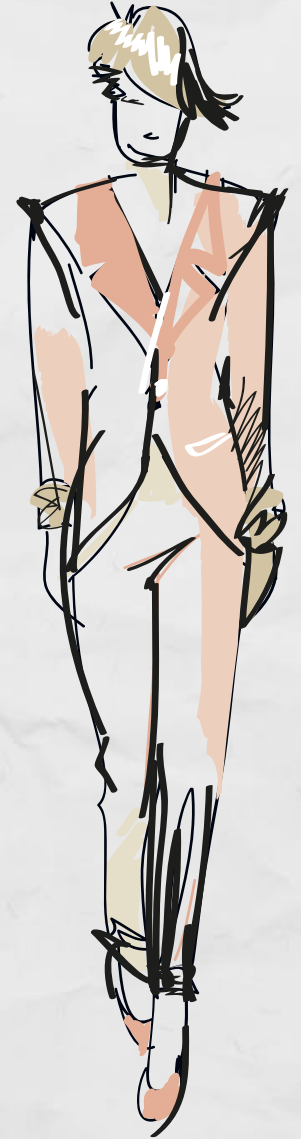
CLUSTERING



**ASSOCIATION RULE
MINING**

Association Rule Mining

- Data mining technique that aims to discover interesting relationships, patterns, and correlations within large datasets.
- It consists of an antecedent (if part) and a consequent (then part). The dataset contains an antecedent, and we derive a consequent by using the antecedent.



Key Concepts

Support : the proportion of transactions in the dataset that contain a specific itemset

$$\text{Support} = \frac{\text{Frequency}(X,Y)}{N}$$

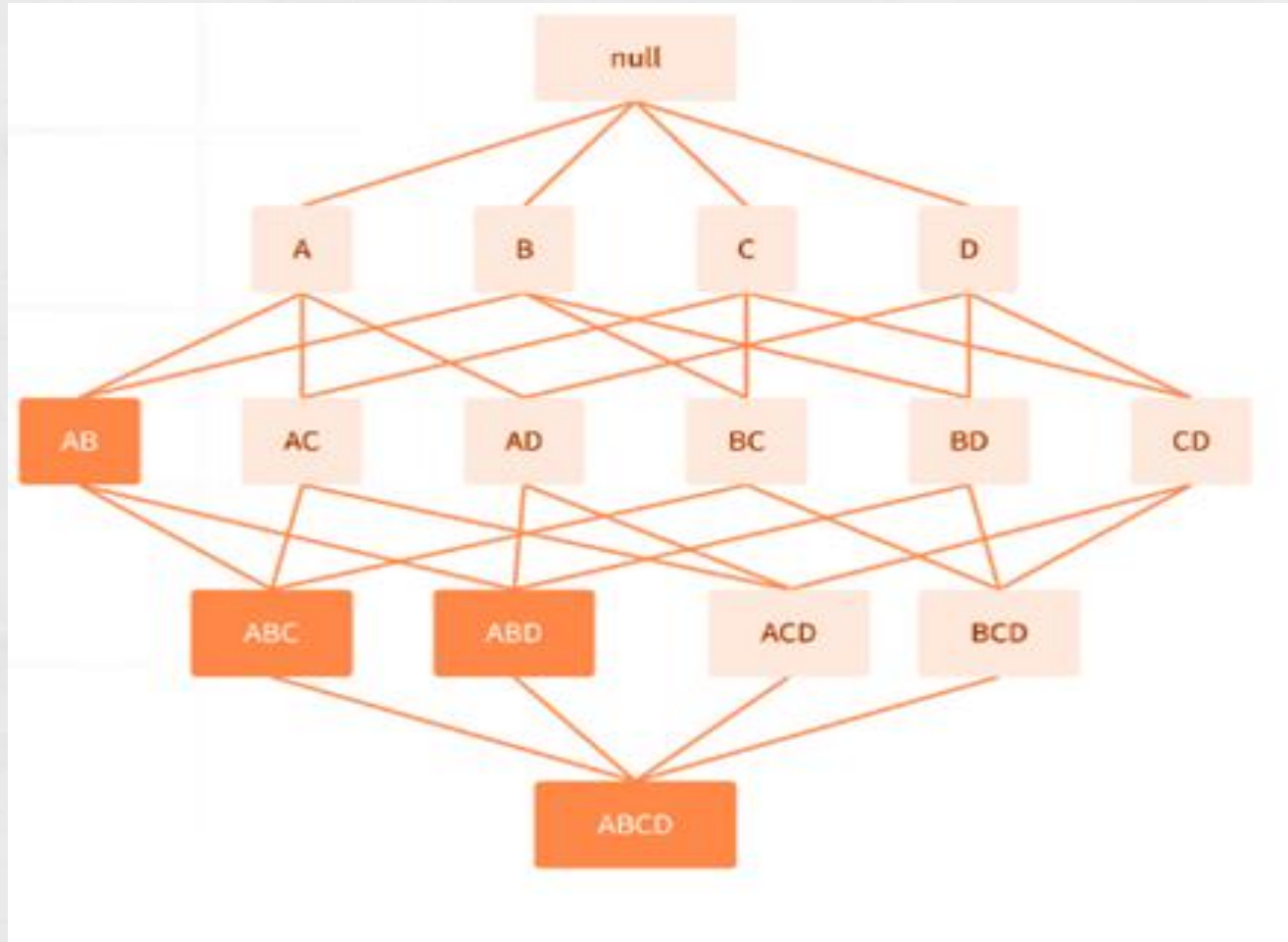
Confidence: the ratio of the number of transactions containing both the antecedent and the consequent to the number of transactions containing only the antecedent.

$$\text{Confidence} = \frac{\text{Frequency}(X,Y)}{\text{Frequency}(X)}$$

Lift: quantifies how likely the consequent is to occur when the antecedent is present compared to when the two events are independent.

$$\text{Lift} = \frac{\text{Support}}{\text{Support}(X) * \text{Support}(Y)}$$

Apriori algorithm



Apriori algorithm

Set Min Support Threshold

Identify Frequent Items

Generate Candidate Itemsets (Size 2)

Prune Infrequent Itemsets

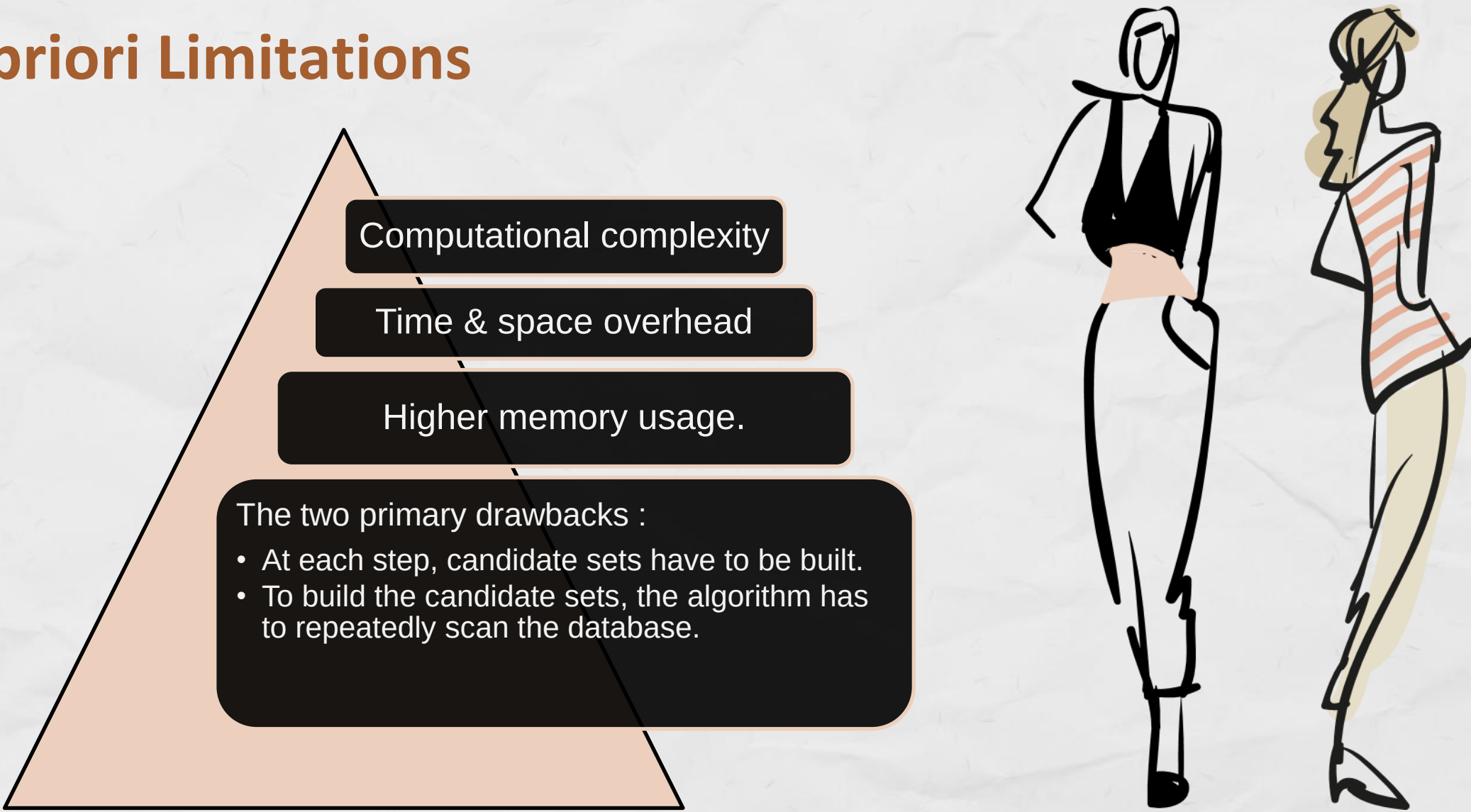
Generate Larger Itemsets

Repeat Pruning

Iterate

Generate Association Rules

Apriori Limitations



Computational complexity

Time & space overhead

Higher memory usage.

The two primary drawbacks :

- At each step, candidate sets have to be built.
- To build the candidate sets, the algorithm has to repeatedly scan the database.

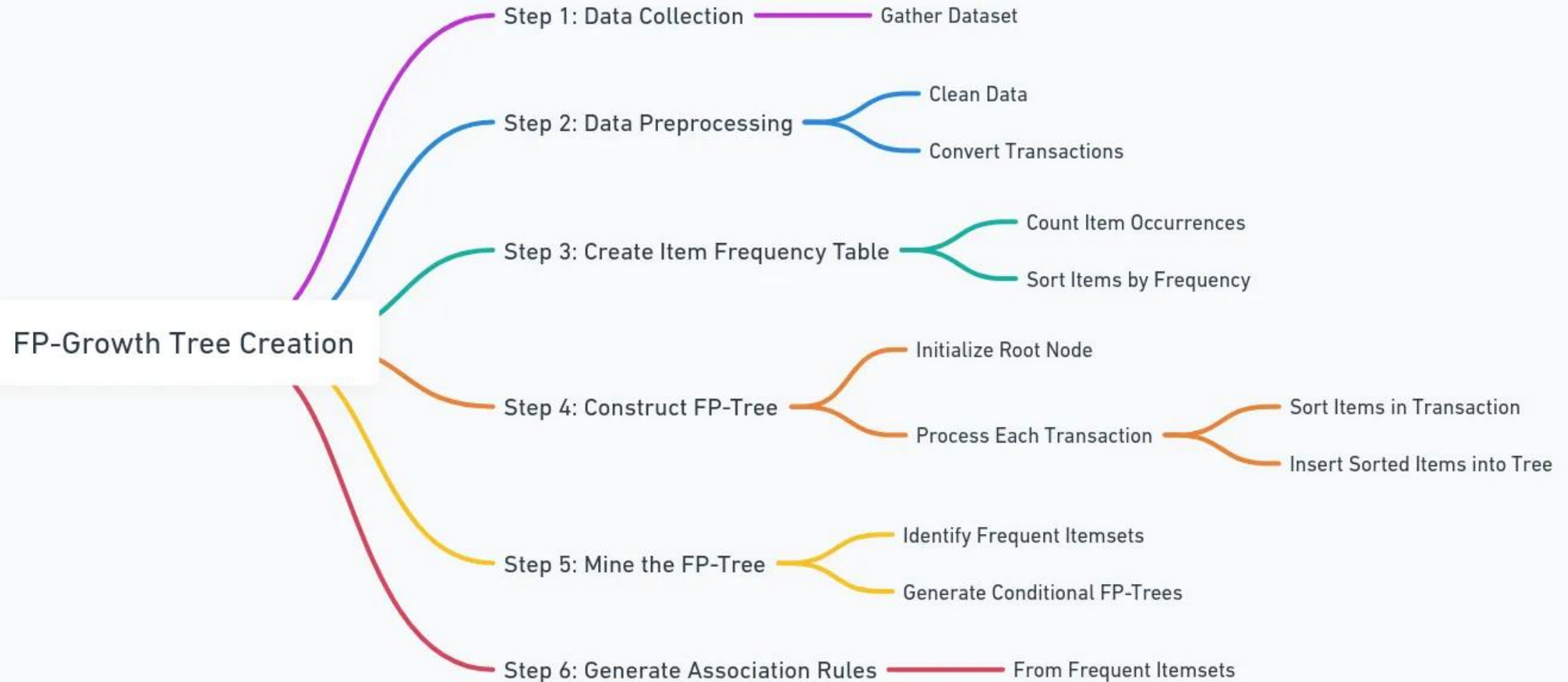
FP-Growth Algorithm

This method is used to mine frequent itemsets within a database without generating candidate sets explicitly.

It uses a compact data structure called the FP-tree (Frequent Pattern Tree) to encode transactions and discover frequent itemsets efficiently.

Transaction ID	Items Purchased
T1	T-shirt, Jeans
T2	Dress, T-shirt
T3	Jeans, Jacket, Sneakers
T4	T-shirt, Jeans, Jacket

FP-Growth Algorithm Steps

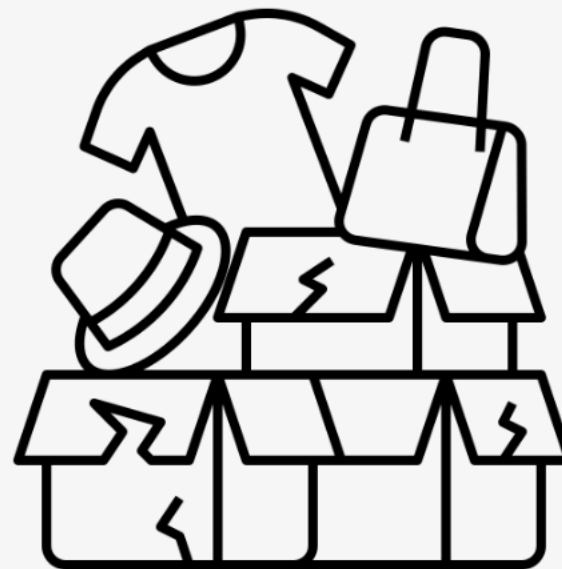


Top Picks Together

Select Product Title:

minimalist hobo bag mini twist ... x ▾

Get Recommendations

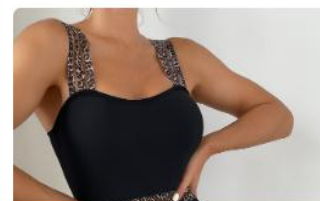


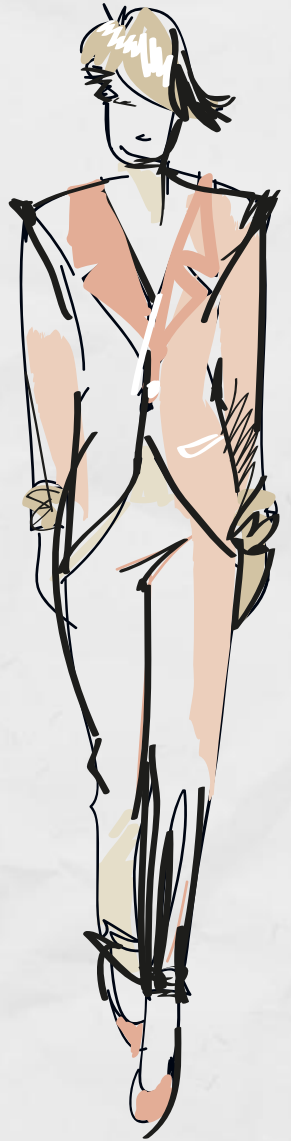
Get Recommendations

Selected Product Image



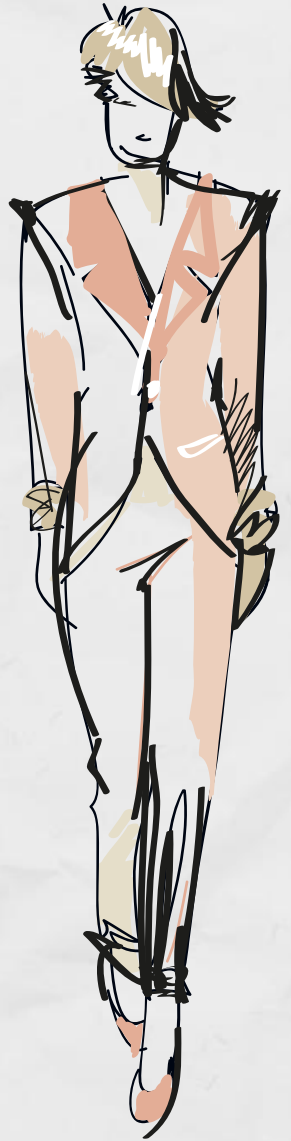
This product is recommended with your selected item because 17 user(s) bought both together.





Similarity Model





Content based



Content type

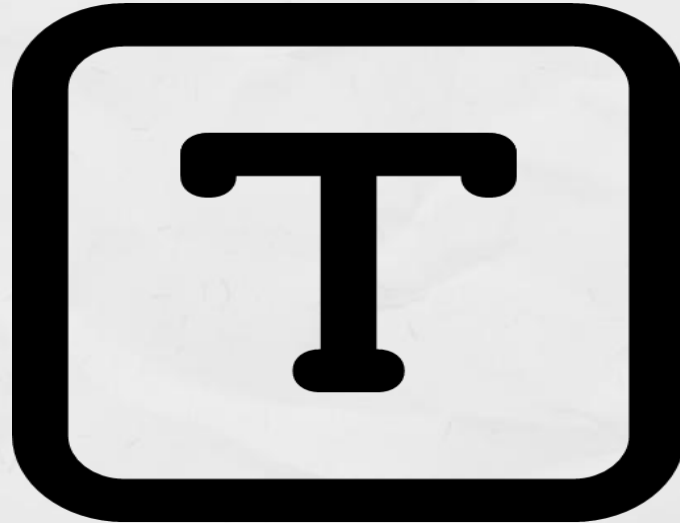
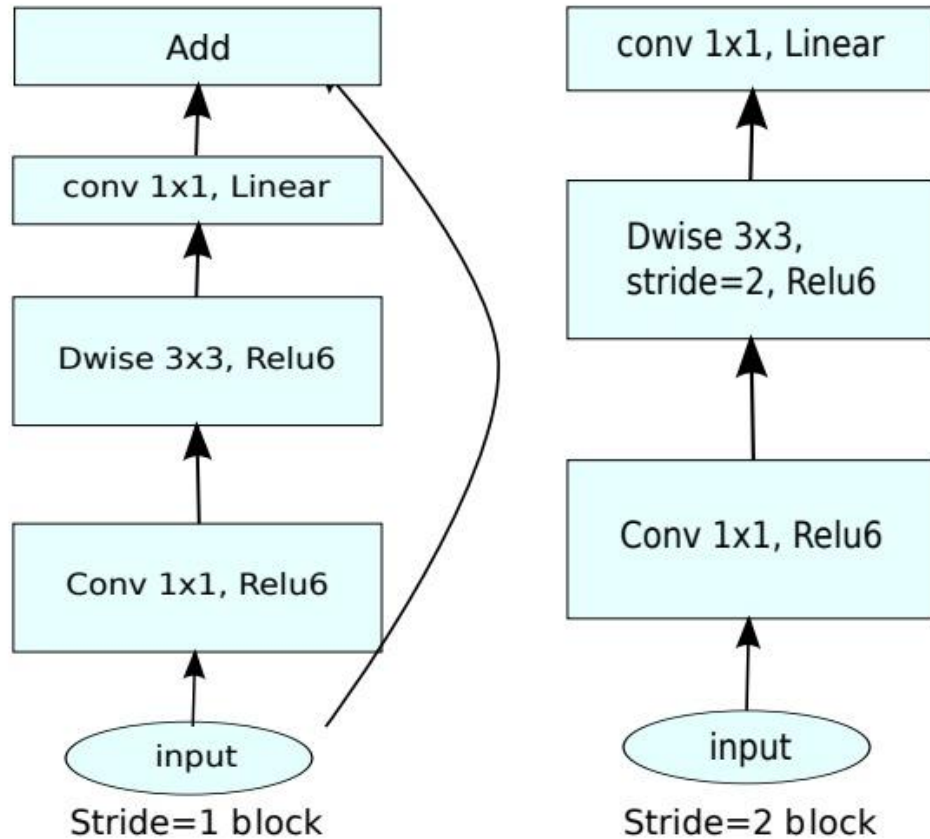




Image Based

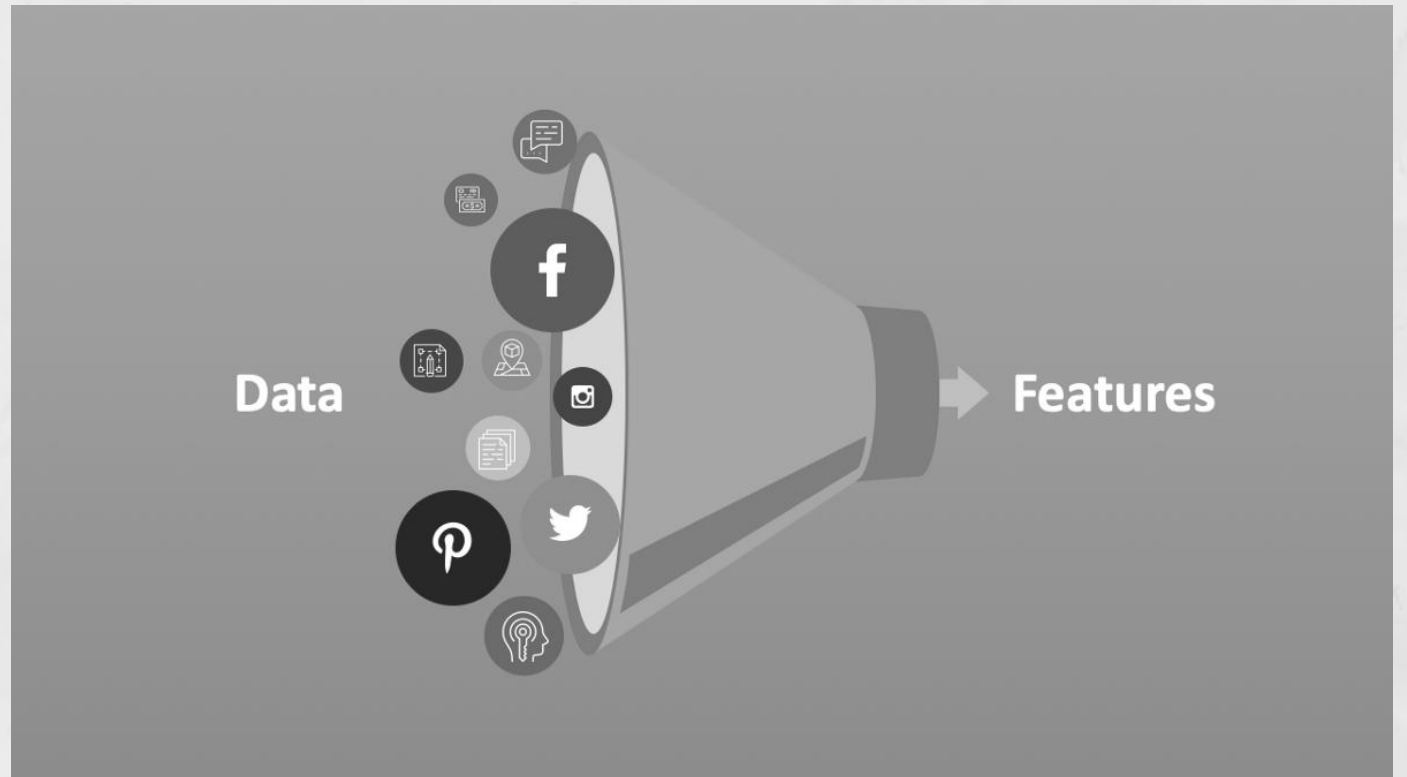
MobileNetV2



Input	Operator	t	c	n	s
$224^2 \times 3$	conv2d	-	32	1	2
$112^2 \times 32$	bottleneck	1	16	1	1
$112^2 \times 16$	bottleneck	6	24	2	2
$56^2 \times 24$	bottleneck	6	32	3	2
$28^2 \times 32$	bottleneck	6	64	4	2
$14^2 \times 64$	bottleneck	6	96	3	1
$14^2 \times 96$	bottleneck	6	160	3	2
$7^2 \times 160$	bottleneck	6	320	1	1
$7^2 \times 320$	conv2d 1x1	-	1280	1	1
$7^2 \times 1280$	avgpool 7x7	-	-	1	-
$1 \times 1 \times 1280$	conv2d 1x1	-	k	-	-

Feature Extraction Process

- Image Fetching and Preprocessing
- Feature Extraction using MobileNetV2
- Storing Features



Similarity Computation

- **Cosine Similarity**

- Top-K Similar Images

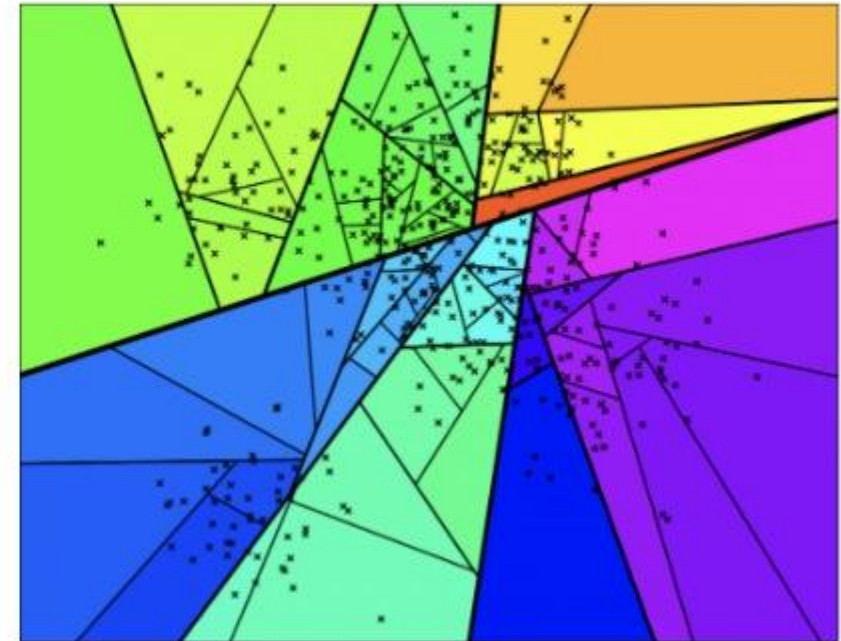
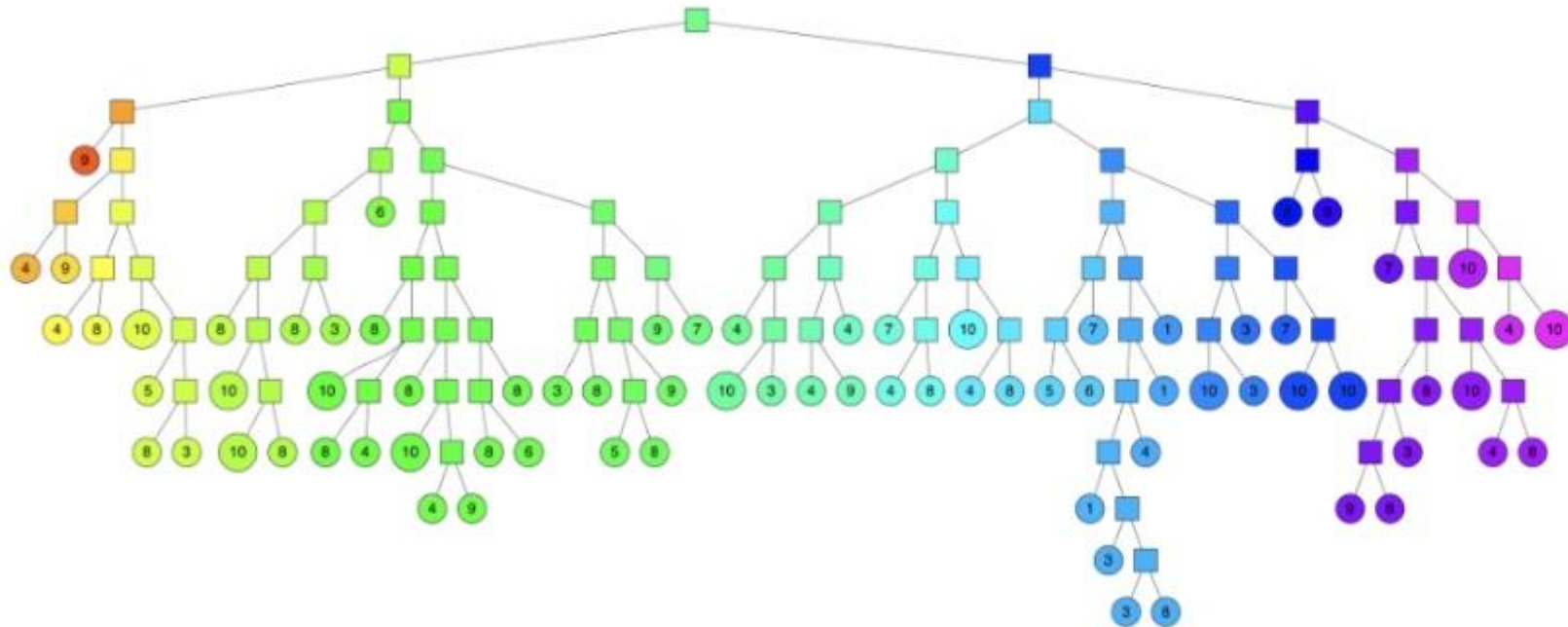
- Retrieving Features
- Calculating Similarities
- Sorting and Selecting

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$



Text Based

Annoy (Approximate Nearest Neighbors Oh Yeah) Index



Feature Extraction using Tf-Idf Vectorization

- Combined Important Features (Color, Target Audience, Department, Product Description)
- Applied Tf-Idf Vectorization on combined features

$$w_{x,y} = tf_{x,y} \times \log \left(\frac{N}{df_x} \right)$$

TF-IDF

Term x within document y

$tf_{x,y}$ = frequency of x in y

df_x = number of documents containing x

N = total number of documents

Using Annoy Index for Similarity Search

**Building the
Annoy Index
based on the Tf-
Idf Matrix**

**Cosine distance
metrics for
similarity search**

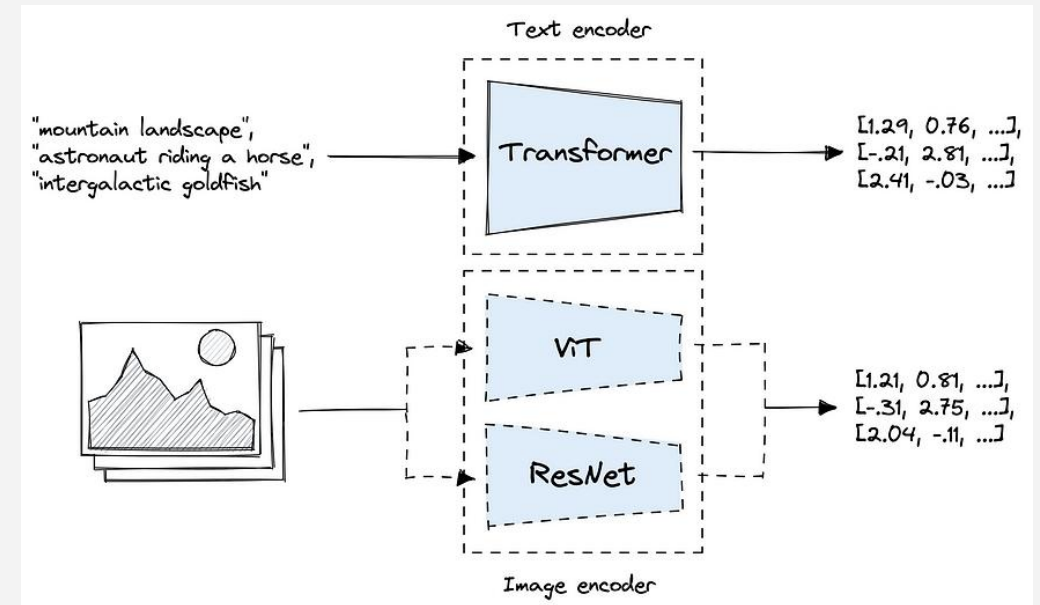


Both
Text & Image

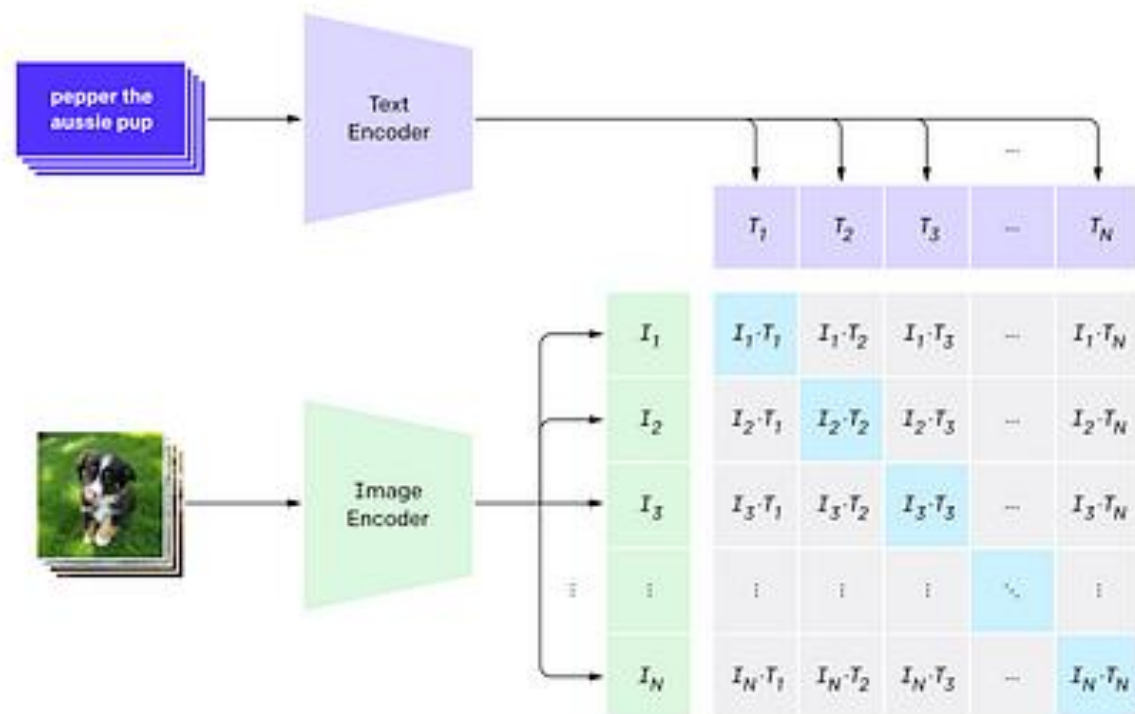
Fashion CLIP

Architecture Overview:

- **Vision Encoder:** Converts images into fixed-size embeddings using a convolutional neural network (Vision Transformer).
- **Text Encoder:** Converts text descriptions into fixed-size embeddings using a transformer-based model (Transformer).

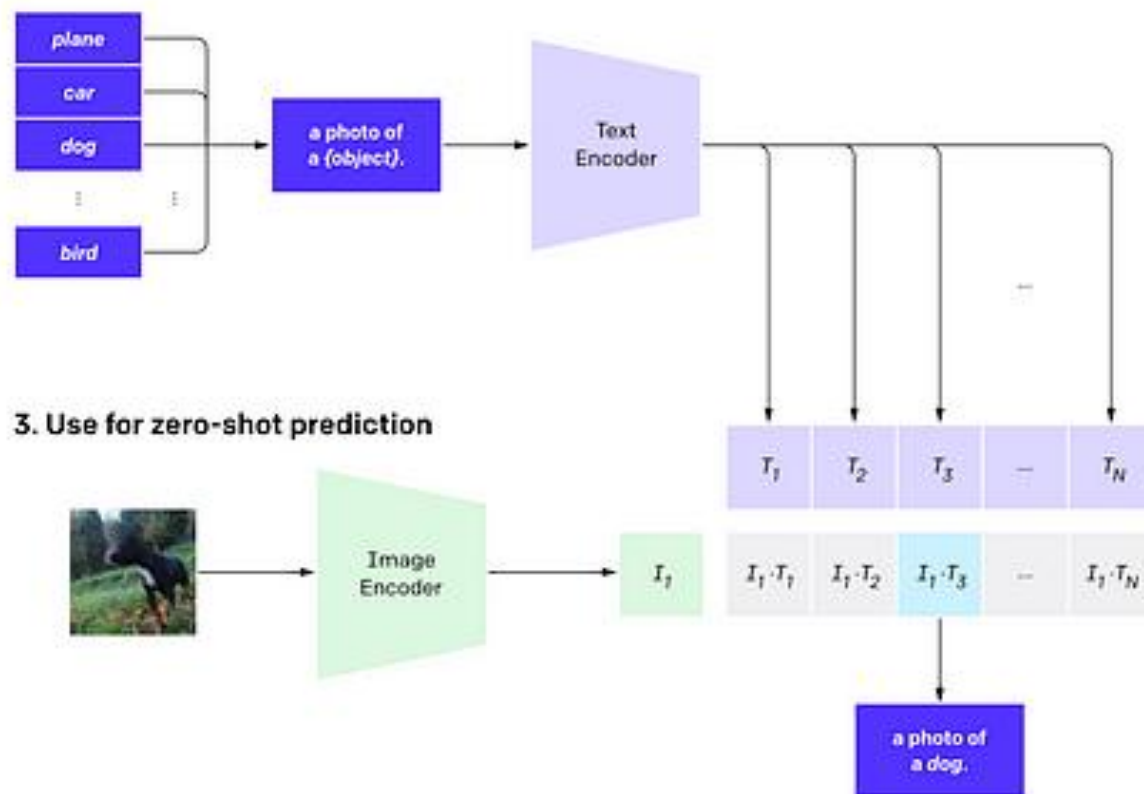


1. Contrastive pre-training



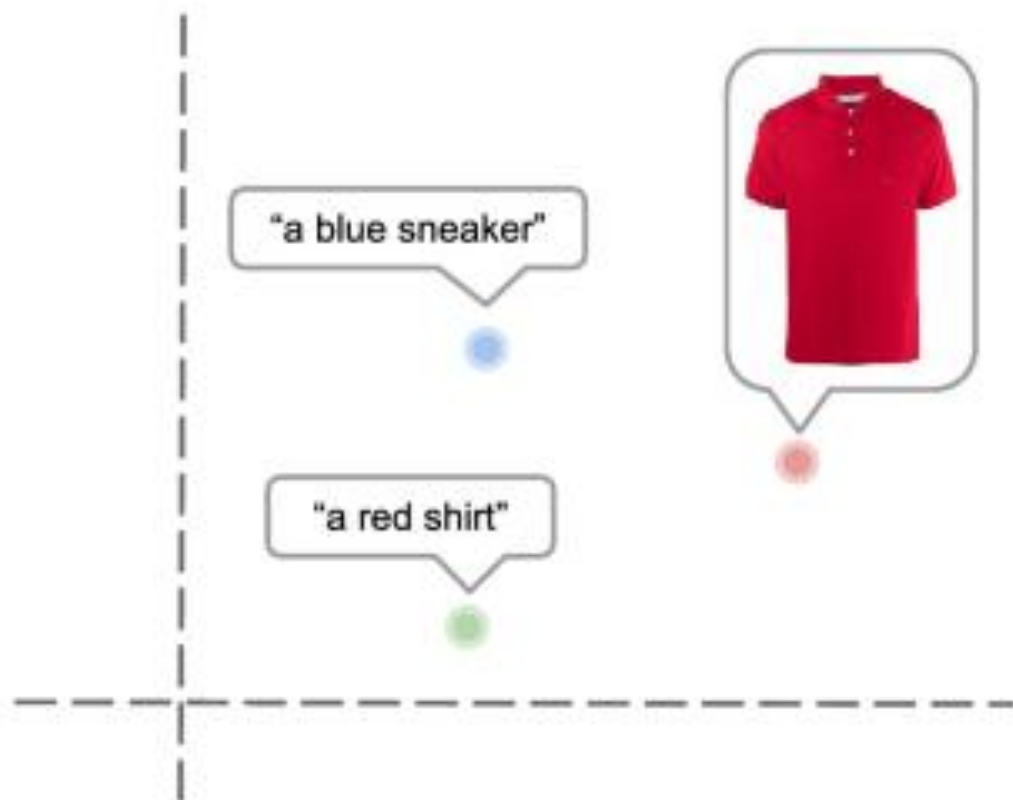
CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset. We then use this behavior to turn CLIP into a zero-shot classifier. We convert all of a dataset's classes into captions such as "a photo of a dog" and predict the class of the caption CLIP estimates best pairs with a given image.

2. Create dataset classifier from label text

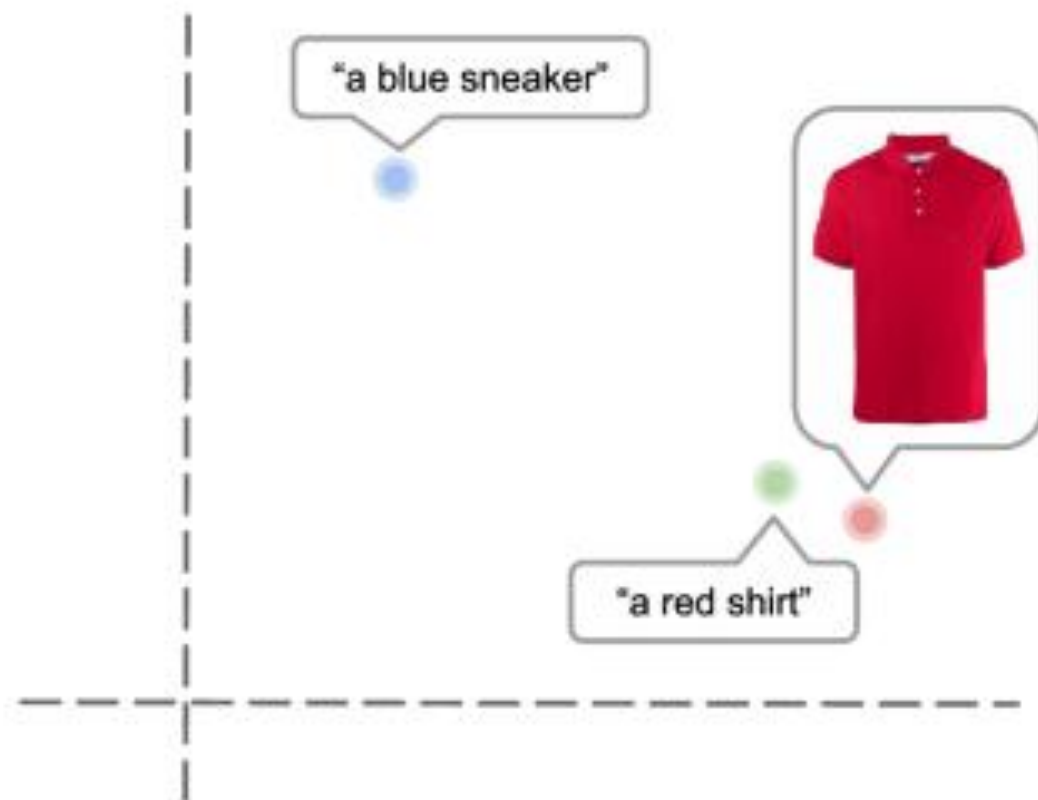


3. Use for zero-shot prediction

Latent Space Before Training



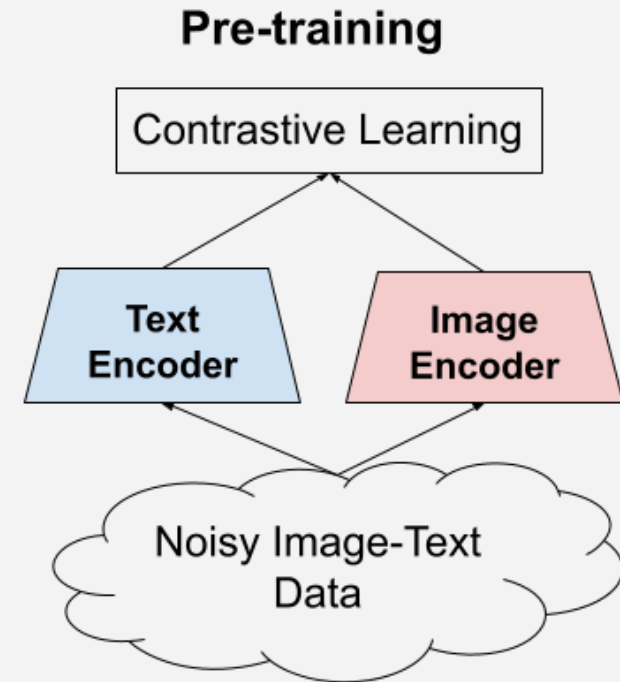
Latent Space After Training



ALIGN

Architecture Overview:

- **Dual-Encoder System:** Separate encoders for images and text.
- **Encoders Used:** EfficientNet for images, BERT for text.
- **Contrastive Loss:** Aligns visual and textual representations in a shared latent space.
- **Training Data:** Over one billion noisy image-text pairs, minimally filtered to retain scale.



Evaluating Retrieval

Setting	Model	Flickr30K (1K test set) R@1		MS-COCO (5K test set) R@1	
		image → text	text → image	image → text	text → image
Zero-shot	ImageBERT	70.7	54.3	44.0	32.3
	UNITER	83.6	68.7	-	-
	CLIP	88.0	68.7	58.4	37.8
	ALIGN	88.6	75.7	58.6	45.6
Fine-tuned	GPO	88.7	76.1	68.1	52.7
	UNITER	87.3	75.6	65.7	52.9
	ERNIE-ViL	88.1	76.7	-	-
	VILLA	87.9	76.3	-	-
	Oscar	-	-	73.5	57.5
	ALIGN	95.3	84.9	77.0	59.9

Similar Styles

Select Product Title:

minimalist hobo bag mini twist ... x v

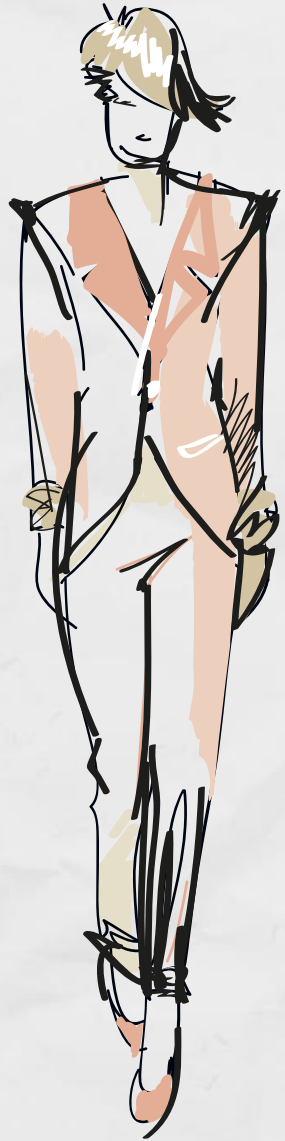
Select Model:

ROBERTA v

ROBERTA

F-CLIP



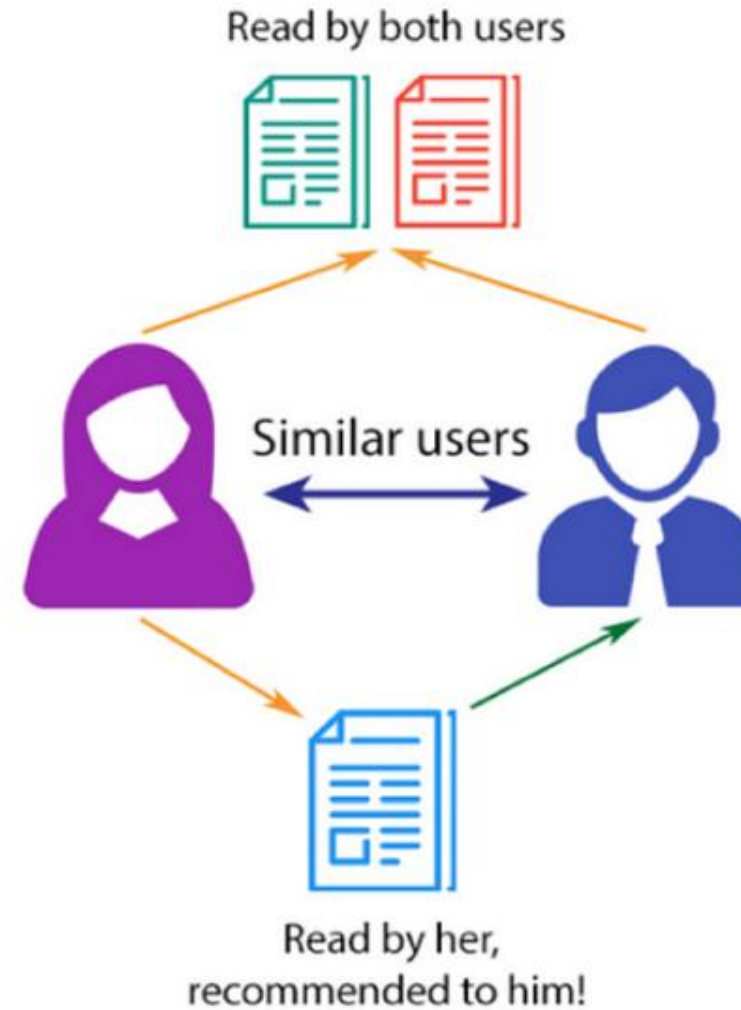


User Based Collaborative Filtering



Collaborative Filtering

COLLABORATIVE FILTERING



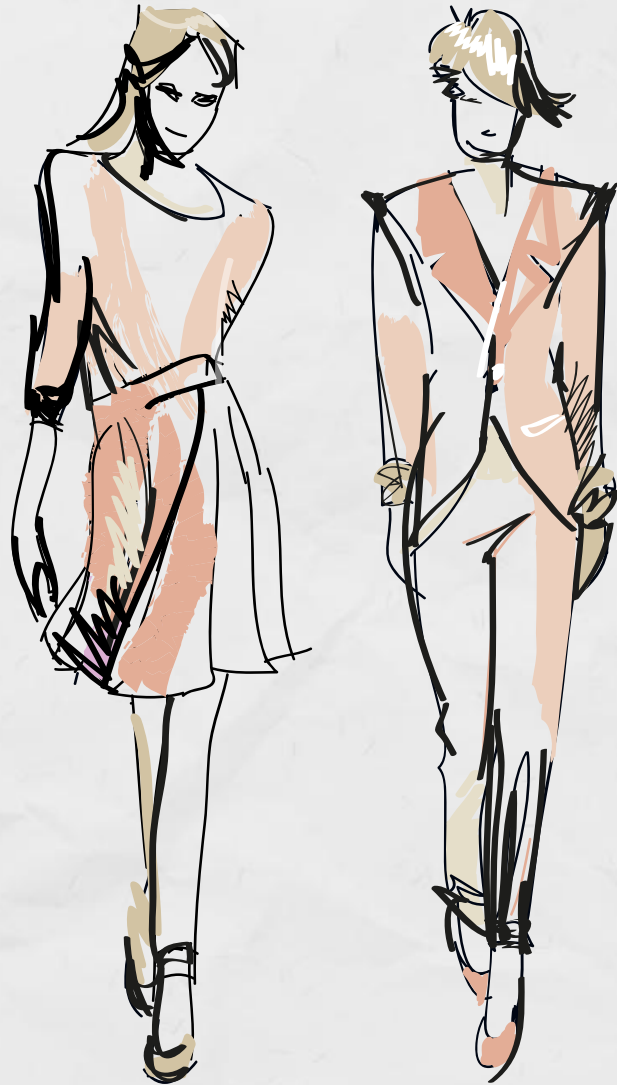
Methods for User Personalization

**Similarity Based on
interaction**

**Similarity Based on
Embeddings**

Two-Tower Model





Similarity Based on Interaction

Overview



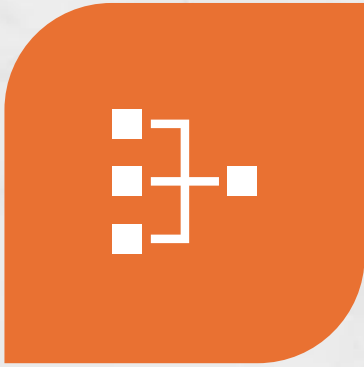
IDENTIFIES USERS WITH SIMILAR
INTERACTION PATTERNS.



USES INTERACTION DATA TO
RECOMMEND ITEMS.



Components



INTERACTION DATA



**SIMILARITY
CALCULATION**



**RECOMMENDATION
GENERATION**

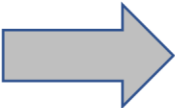


Items



	10	-1	8	10	9	4
	8	9	10	-1	-1	8
	10	5	4	9	-1	-1
	9	10	-1	-1	-1	3
	6	-1	-1	-1	8	10

Users

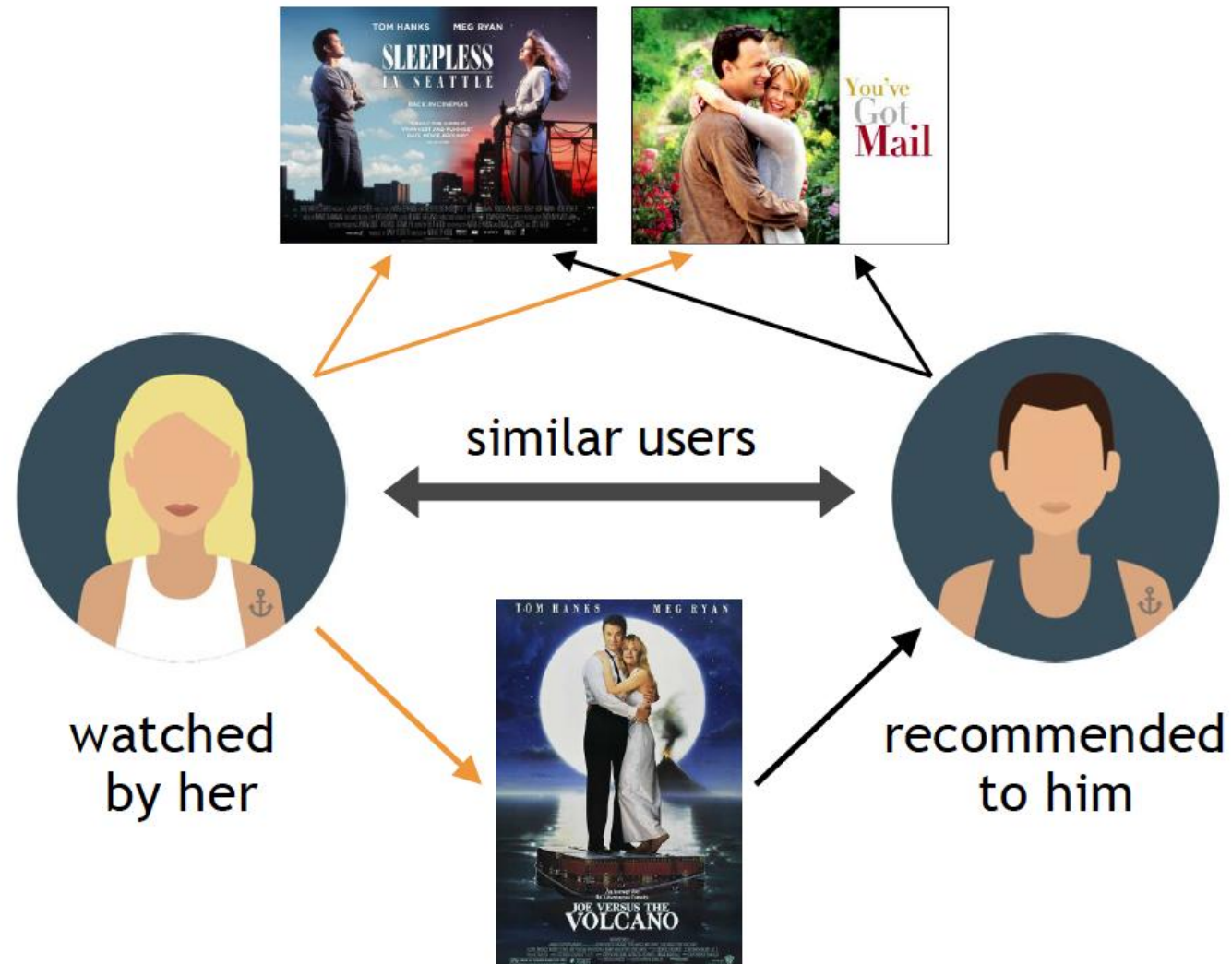


User-item
Interaction
matrix

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Collaborative Filtering

watched by both users





Similarity Based on Embeddings

Overview

Uses embeddings
to capture
relationships in a
high-dimensional
space.

Computes similarity
between user
embeddings for
recommendations



Components

Embeddings Generation

User Embeddings Calculation

Similarity Calculation

Recommendation Generation

Embedding generation

- CLIP Fashion can be used to generate embeddings for items based on their descriptions or titles
- Compute user embeddings as the average of the embeddings of the items they have interacted with.

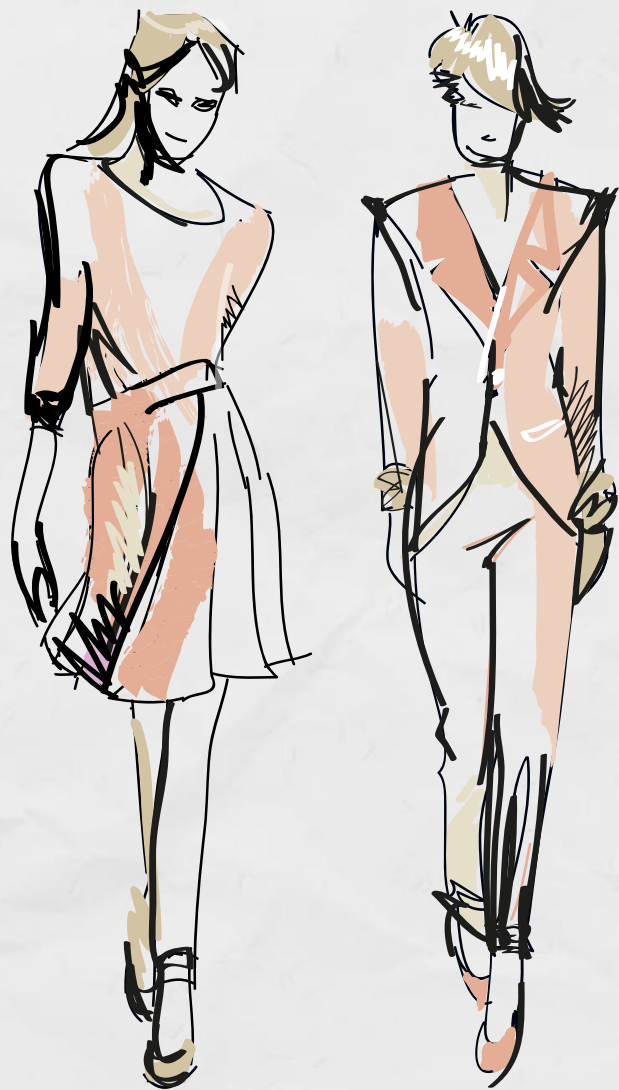


Similarities



Recommendation Generation





Tow Tower
—

Components



USER TOWER



ITEM TOWER



INTERACTION LAYER

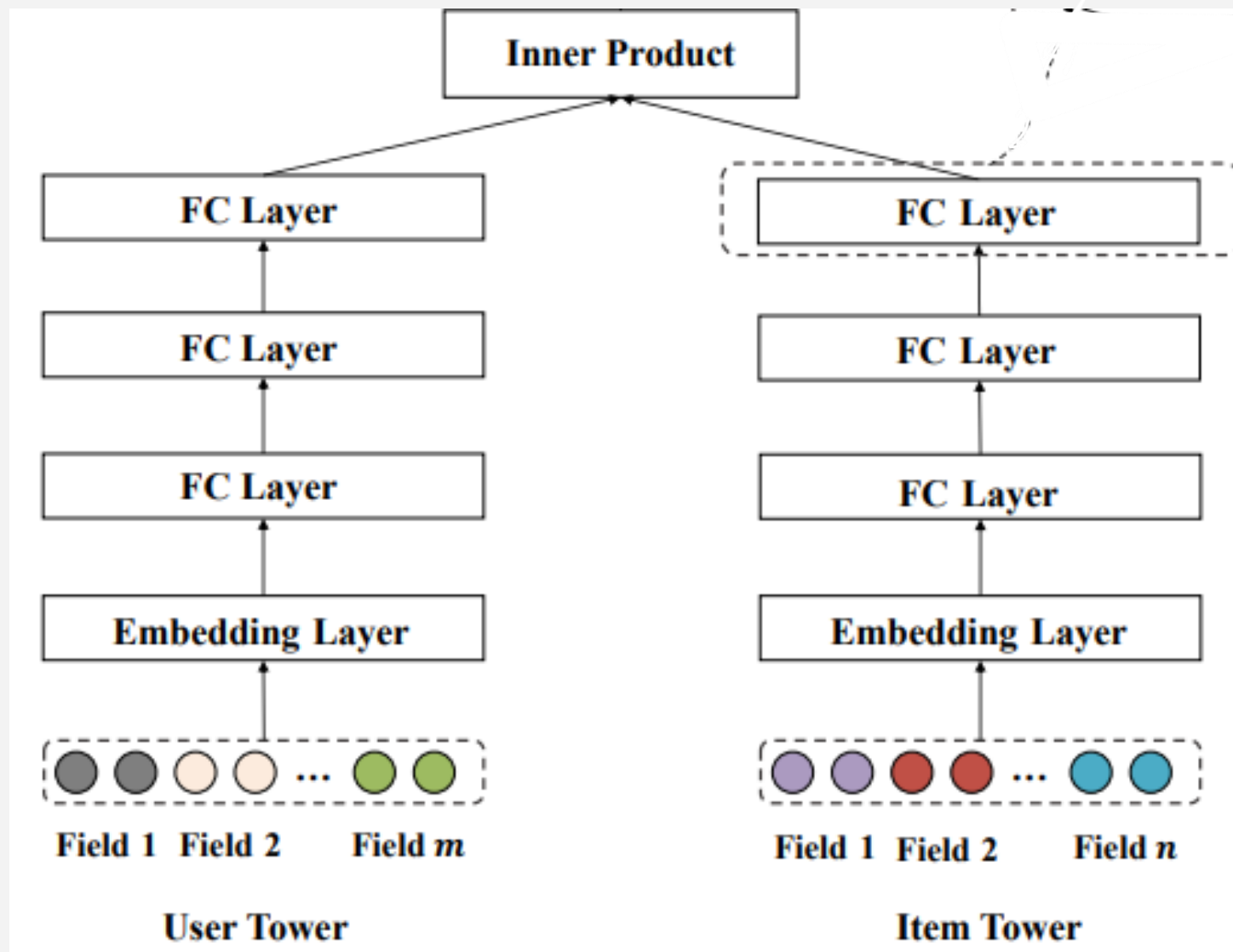


TRAINING



**RECOMMENDATION
GENERATION**

Two-Tower Model





Choosing the Right Approach

Personalized Picks

Sign Up

Login

Select a user ▼

Get Recommendations



Documentation



Scan The Code

AI-Vision

THANKS

